

# Communication Systems (ENEX 351)

## Organized Question Bank

Compiled from Past Year Question Papers

### Contents

|          |                                                                                                                                    |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                                                                                | <b>2</b>  |
|          | Review of Signals and Systems . . . . .                                                                                            | 2         |
|          | Block Diagram of Analog and Digital Communication Systems . . . . .                                                                | 5         |
|          | System Needs and Requirements . . . . .                                                                                            | 6         |
|          | Noise, Attenuation, and Interference . . . . .                                                                                     | 6         |
| <b>2</b> | <b>Amplitude Modulation</b>                                                                                                        | <b>7</b>  |
|          | Time and Frequency Domain Expressions, Modulation Index, Bandwidth . . . . .                                                       | 7         |
|          | AM for Single and Double Tone Message, Carrier and Sideband, Power, Bandwidth<br>and Power Efficiency, Hilbert Transform . . . . . | 7         |
|          | DSB-FC: Generation (Square Law) and Detection . . . . .                                                                            | 8         |
|          | DSB-SC: Generation (Linear/Balanced Modulator), Synchronous Detection . . . . .                                                    | 8         |
|          | Overview of SSB, VSB, and ISB . . . . .                                                                                            | 8         |
|          | Phase Locked Loop (PLL), AM Demodulation using PLL . . . . .                                                                       | 8         |
|          | Super-heterodyne AM Receiver . . . . .                                                                                             | 8         |
| <b>3</b> | <b>Angle Modulation</b>                                                                                                            | <b>9</b>  |
|          | Basic Definitions, Time Domain Expressions for FM and PM . . . . .                                                                 | 9         |
|          | Time Domain Expression for Single Tone Modulated FM Signals, Spectral Represen-<br>tation . . . . .                                | 9         |
|          | Bandwidth of FM, Carson's Rule, Narrow and Wideband FM . . . . .                                                                   | 9         |
|          | Generation of FM: Direct and Indirect . . . . .                                                                                    | 9         |
|          | Demodulation of FM: Non-synchronous and Synchronous . . . . .                                                                      | 9         |
|          | Stereo FM, Pre-emphasis and De-emphasis Network . . . . .                                                                          | 9         |
|          | Super-heterodyne Radio Receiver for FM . . . . .                                                                                   | 9         |
| <b>4</b> | <b>Pulse Modulation</b>                                                                                                            | <b>10</b> |
|          | Sampling Theorem, Ideal/Practical Sampling, Aliasing, Aperture Effect, Reconstruction                                              | 10        |
|          | Fundamentals of PAM, PWM, PPM . . . . .                                                                                            | 10        |
|          | PCM: Quantization, Quantization Error and Noise . . . . .                                                                          | 10        |
|          | SQNR in Uniform Quantization, Non-uniform Quantization, Companding . . . . .                                                       | 11        |
|          | DPCM, DM: Encoder, Decoder, Advantages, Disadvantages, Noise in DM . . . . .                                                       | 11        |
| <b>5</b> | <b>Multiplexing Techniques</b>                                                                                                     | <b>11</b> |
|          | Multiplexing Fundamentals: FDM, TDM, WDM and Applications . . . . .                                                                | 11        |
|          | T1 and E1 TDM PCM Telephony Hierarchy . . . . .                                                                                    | 12        |
|          | Multiple Access Fundamentals: FDMA, TDMA, CDMA, SDMA . . . . .                                                                     | 13        |

|          |                                                                                                      |           |
|----------|------------------------------------------------------------------------------------------------------|-----------|
| <b>6</b> | <b>Baseband Digital Data Transmission</b>                                                            | <b>13</b> |
|          | Information Theory, Entropy, Symbol Rates and Data Rates . . . . .                                   | 13        |
|          | Shannon-Hartley Channel Capacity Theorem . . . . .                                                   | 14        |
|          | Compression Techniques: Shannon-Fano, Huffman Codes . . . . .                                        | 14        |
|          | Line Coding Schemes: Unipolar, Polar, Bipolar . . . . .                                              | 16        |
|          | RZ, NRZ, AMI, Manchester, Differential Manchester, B8ZS, HDB3 . . . . .                              | 16        |
|          | ISI, Nyquist Criteria, Pulse Shaping for Zero ISI . . . . .                                          | 17        |
| <b>7</b> | <b>Digital Modulation Techniques</b>                                                                 | <b>17</b> |
|          | Binary Digital Modulation (ASK, FSK, PSK) . . . . .                                                  | 17        |
|          | QPSK Generation, Properties, Constellation Diagram, Detection . . . . .                              | 18        |
|          | M-ary Modulation Techniques, M-PSK versus M-QAM . . . . .                                            | 18        |
| <b>8</b> | <b>Error Detection and Correction Coding</b>                                                         | <b>18</b> |
|          | Hamming Weight, Hamming Distance, Code Vectors, Constraint Length, Code Rate,<br>Syndromes . . . . . | 18        |
|          | Error Detection Codes: Checksum, CRC . . . . .                                                       | 19        |
|          | Error Correction Codes: Linear Block Codes, Hamming Codes . . . . .                                  | 19        |
|          | Cyclic Codes (Generator Polynomial, Parity-Check Polynomial) . . . . .                               | 20        |
| <b>9</b> | <b>Noise in Communication Systems</b>                                                                | <b>20</b> |
|          | Definition, White Noise, AWGN Channel, PSDF, Autocorrelation of White Noise . . .                    | 20        |
|          | Ideal Low-pass and RC Filtering of White Noise, Noise Equivalent Bandwidth . . . .                   | 21        |
|          | Optimum Detection, Matched Filter, Realization (Time Correlators) . . . . .                          | 21        |
|          | Overview of Error Probability Function in Digital Communication (ASK, FSK, PSK)                      | 21        |
| <b>A</b> | <b>Out-of-Syllabus Questions</b>                                                                     | <b>23</b> |
|          | <b>Out-of-Syllabus Questions</b>                                                                     | <b>23</b> |

# 1 Introduction

## Review of Signals and Systems

1. Differentiate data and signal with two examples of each. [2078 Kartik — CT602]
2. Define periodic and non-periodic signals with examples. [2078 Kartik — CT602]
3. Test the stability of the system  $h(t) = e^{4t}u(t)$ . [2078 Kartik — CT602]
4. Test the given function  $y(t) = t.x(t)$  for causality, non causality and anti causality. [2078 Kartik — CT602]
5. Differentiate between energy and power signals with examples. Determine if the following system is stable LTI:

$$Y(t) = x(t^2)$$

[2078 Bhadra — CT602]

6. Explain how we plot line spectrums of a continuous time signal and illustrate an example. [2078 Bhadra — CT602]
7. Define a causal system and a stable system. Check whether a system  $y(t) = x(t) + x(t-1)$  is (i) linear (ii) time-invariant (iii) causal (iv) memoryless system. [2079 Bhadra — CT602]
8. State properties of continuous time Fourier Transform. [2079 Bhadra, 2080 Baishakh — CT602]
9. Find convolution between two signals  $x_1(t) = u(t) - u(t-3)$  and  $x_2(t) = u(t+4) - u(t)$ . [2079 Bhadra — CT602]
10. Determine whether or not the following CT signal is periodic. If the signal is periodic, determine its fundamental period.  $x(t) = 3 \cos(4t + \pi/3)$ . [2080 Baishakh — CT602]
11. Test the linearity of the systems,  $y(t) = x^2(t)$  and  $y(t) = 3x(t) + 5$ . [2080 Baishakh — CT602]
12. Test the systems static or dynamic,  $y(t) = 2x(t)$  and  $y(t) = 2x(t) - 3x(t-3)$ . [2080 Baishakh — CT602]
13. Find the output response of an LTI system when input  $x(t) = e^{-t}u(t)$  and impulse response  $h(t) = e^{-2t}u(t)$ . [2080 Baishakh — CT602]
14. Find magnitude and phase spectrum of  $x(t) = 4 + 2 \cos \frac{2\pi}{3}t + 4 \sin \frac{5\pi}{3}t$ . [2080 Baishakh — CT602]
15. Discuss any four properties of Fourier Transform. [2080 Baishakh — CT602]
16. Define energy and power signal with examples. Check whether the following system is linear, causal and time invariant or not:  $y(t) = \log x(t)$ . [2081 Baishakh, 2082 Kartik — EX656]
17. Define a linear system and a time invariant system. Evaluate the odd and even component of a signal defined as  $x(t) = e^{-2t}u(t)$ . [2081 Bhadra — CT602]
18. Prepare a signal flow diagram of a system that combines at least seven mathematical operations in an equation. [2081 Bhadra, 2082 Baishakh — CT602]

19. Compare and contrast Fourier series and Fourier transform. Mention the conditions and properties of Fourier transformation. [2081 Bhadra — CT602]
20. Define energy and power signal. Evaluate the power or energy of the following signals: (a)  $x(t) = e^{-2|t|}$ , (b)  $x(t) = e^{j0.1\pi t}$ . [2082 Baishakh — CT602]
21. Mention the characteristics of a Linear Time-Invariant (LTI) system. How do you test the stability of a system? Test the stability of the following systems: (i)  $y(t) = x^2(t)$ , (ii)  $y[t] = 50 \sin(x(t)) + 10$ . [2082 Baishakh — CT602]
22. Write basic properties of systems. A system's output is given as  $z(t) = t^2 y(t - 1)$ . Determine whether the system is Linear or not. [2082 Bhadra — CT602]
23. Classify the following signals whether it is energy/power, continuous/discrete signal or neither: (i)  $e^{-t}u(t)$ , (ii)  $\delta[n - 2]$ . [2082 Bhadra — CT602]
24. List the property of LTI system. Prove frequency shifting properties of continuous time Fourier series. [2082 Bhadra — CT602]
25. Define the unit impulse function and the unit step function. Find the Fourier transform of a unit impulse signal in continuous time domain. [2082 Bhadra — CT602]
26. Find the odd and even symmetry of the given signals graphically (triangular and right-triangular pulses of amplitude 2 spanning  $t = 0$  to 2).

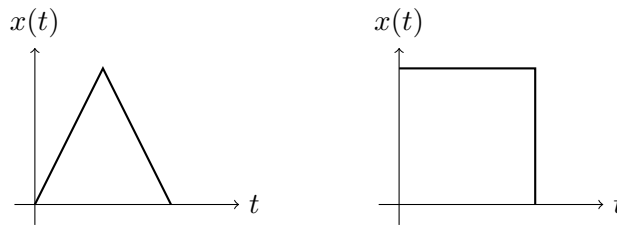


Figure 1: \*  
Signals for odd/even symmetry evaluation (reconstructed)

[2083 Baishakh — CT602]

27. With suitable examples, explain how time invariance of a system is checked. [2082 Bhadra — EX656]
28. With suitable examples, explain how linearity of a system is checked. [2083 Baishakh — EX656]
29. Mention the conditions and properties of Fourier transformation. [2083 Baishakh — EX656]
30. Plot the magnitude and phase spectrums of  $x(t) = 6 - 8 \sin(2\pi 160t + 30^\circ) - 6 \cos(\pi 160t - 50^\circ) + 2 \sin(300\pi t + 120^\circ)$ . [2082 Bhadra — ENCT253]
31. Using suitable examples, explain how time invariance of a system is checked (with real-world half/full-duplex context). [2082 Bhadra — ENCT253]
32. What is distortionless transmission? Differentiate energy and power signal. Describe the impulse response of LPF. [2079 Ashwin — EX652]

33. What is autocorrelation and cross correlation of a signal? Explain and derive power spectrum density function of a signal. [2079 Ashwin — EX652]
34. Determine the odd and even component of a signal  $x(t) = e^{-2t}u(t)$ . [2081 Bhadra — CT602]
35. State properties of continuous time Fourier Transform. Convolve the following signals graphically (rectangular pulse and rectangular pulse of amplitude 1).

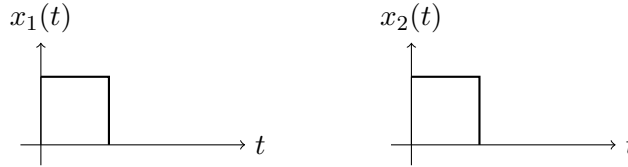


Figure 2: \*  
Signals to be convolved graphically (reconstructed)

[2081 Baishakh, 2081 Bhadra — CT602]

36. Define LTI system. Convolve the following signals graphically:

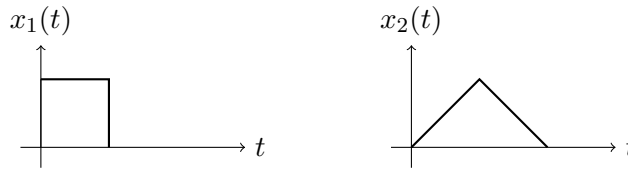
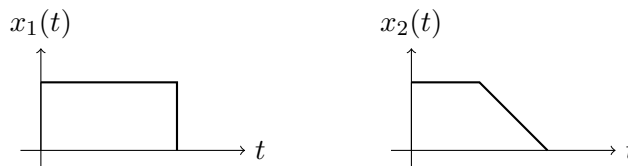


Figure 3: \*  
Rectangular and triangular pulse to be convolved graphically

[2080 Bhadra — CT602]

37. Determine whether the signal  $y(t) = 2x(2t) + 3x(t - 3) + 4$  is time invariant or not. Consider any two signals and convolve them graphically. [2082 Chaitra — EX656]
38. Perform convolution of the following signals by graphical method, and construct the signal flow diagram of the system described by:

$$y(t) + 2 \int y(t) dt + 3 \iint y(t) dt dt = 4 \int x(t) dt + 5 \iiint x(t) dt dt$$



[2081 Ashwin — EX656]

39. Determine whether the following signals are: (a)  $y(t) = 2t^2x(t) + 3tx(t - 4)$  linear or not, (b)  $y(t) = x^2(t) + x(t + 2)$  causal or not. [2081 Chaitra — EX656]

40. Derive auto-correlation, cross-correlation and convolution of the following two signals graphically (rectangular and triangular pulses of amplitude 2, spanning  $t = 1$  to 2). [2081 Chaitra — EX656]
41. Determine whether the following signals are: (a)  $y(t) = 2t^2x(t) + 3tx(t - 4)$  linear or not, (b)  $y(t) = x^2(t) + x(t + 2)$  causal or not (repeated with 2082 Chaitra data set). [2082 Chaitra — EX656]
42. Taking examples of any two signals, derive auto-correlation, cross-correlation and convolution graphically. [2082 Kartik — EX656]

### Block Diagram of Analog and Digital Communication Systems

43. Explain the procedure of converting an analog signal to digital. Also, briefly explain each step involved. [2078 Kartik — CT602]
44. Define Data and Signal with examples. Sketch a generic block diagram of a digital data communication system for full duplex mode. [2079 Bhadra, 2081 Baishakh, 2081 Bhadra — CT602]
45. Define data and signal. Elaborate the advantages of digital communication system as compared to analog communications system. Explain the basic causes of transmission impairments in communication system. [2080 Baishakh — CT602]
46. Prepare a generic block diagram of a digital communication system representing communications between an air traffic controller and an in flight pilot in a cockpit. Explain the basic causes of transmission impairments in communication system. [2081 Bhadra — CT602]
47. Prepare a generic block diagram of a communication system that represents communication between a mobile and a landline telephone user. [2082 Baishakh — CT602]
48. Explain generic block diagram of digital communication system for analog data to be communicated. List the transmission impairments in data communication. [2082 Bhadra — CT602]
49. Using suitable generic real-world examples of digital communication systems, compare the functioning of half- and full-duplex modes. Using a suitable generic real-world example of a digital communication system, explain its functioning in full-duplex mode. [2082 Bhadra, 2083 Baishakh — ENCT253]
50. Write short note on: Analog versus digital communication. [2083 Baishakh — CT602]
51. What is communication system? Explain the importance of transducers, modulators and channel in Analog communication system with neat and clean diagram. [2079 Ashwin — EX652]
52. What are the reasons for modulation? Write the advantages of digital communication over analog communication. Sketch a generic block diagram of a digital communication for full-duplex mode. [2079 Chaitra — EX656]
53. Draw a generic block diagram of Analog communication system for half-duplex mode. Explain the need and disadvantages of modulation. [2080 Ashwin — EX656]
54. Sketch and describe the functional digital communication system working in a duplex mode and contrast it with an analog communication system. Mention its impairments and performance parameters. [2081 Chaitra — EX656]

55. Describe the functional digital communication system working in a half-duplex mode and contrast it with an analog communication system. Mention all the communication impairments and performance parameters. [2082 Kartik — EX656]
56. Sketch a proper generic block diagram of a voice communication system between an on-board pilot and an air traffic controller, and briefly explain the needs and functioning of each building block of that system. [2082 Chaitra — EX656]
57. Define modulation. Draw the functional block diagram of analog communication system and briefly explain each block. [2079 Chaitra — EX652]
58. List the differences between digital communication system and analog communication system. Draw functional block diagrams of DCS and explain the significance of each major block. [2080 Bhadra — EX702]
59. Explain digital communication system with appropriate block diagram. [2080 Baishakh — EX702]
60. Briefly explain the function of the basic components of a digital communication system. [2079 Bhadra — EX702]

### **System Needs and Requirements**

61. List and describe all data communication parameters that describe the performance of the system. [2078 Bhadra — CT602]
62. Explain the basic performance of communication channel in the data communication system. [2082 Bhadra — CT602]
63. Define any five performance parameters that describe the quality of data communication. [2083 Baishakh — CT602]
64. Define transmission performance parameters in a data communication system. [2080 Bhadra, 2081 Baishakh — CT602]
65. Define any ten performance parameters/impairments of digital communication system. [2082 Chaitra — EX656]
66. Define any six performance/impairment parameters common in digital data communication. [2082 Bhadra, 2083 Baishakh — ENCT253]

### **Noise, Attenuation, and Interference**

67. Explain the basic causes of transmission impairments in communication system. [2080 Baishakh, 2081 Bhadra — CT602]
68. Differentiate among noise, interference and distortion. [2078 Chaitra — EX656]
69. Discuss the difference between distortion, noise and interference. [2080 Chaitra — EX652]
70. Define distortion and explain its types. [2081 Baishakh — EX702]
71. Define distortion and explain its types (with examples). [2081 Bhadra — EX702]

## 2 Amplitude Modulation

### Time and Frequency Domain Expressions, Modulation Index, Bandwidth

72. With the help of necessary mathematical expressions and sketches, explain the principle of AM. [2083 Baishakh — CT602]
73. Compare AM and FM with basic mathematical formulations and signal diagrams. [2082 Bhadra — CT602]
74. Compare single tone-AM with that of double-tone with basic mathematical formulations and performance parameters. [2083 Baishakh — ENCT253]
75. Compare A.M and F.M in terms of power efficiency and system complexity. [2080 Baishakh — EX702]
76. With the help of necessary mathematical expressions of Bandwidth and Modulation Index, differentiate between AM and FM. [2081 Ashwin — EX656]

### AM for Single and Double Tone Message, Carrier and Sideband, Power, Bandwidth and Power Efficiency, Hilbert Transform

77. Why we need modulation? A modulating signal,  $m(t) = 10 \cos(2\pi \times 10^3 t)$  is amplitude modulated with carrier signal,  $c(t) = 50 \cos(2\pi \times 10^4 t)$ . Find the modulation index, the carrier power and the power required for transmitting AM wave. [2079 Bhadra — CT602]
78. Consider AM signal as  $m(t) = 5[1 + 2 \cos(2\pi + 1000t)] \cos(2\pi \times 10^8 t)$ . Determine modulation index, total power delivered. [2080 Baishakh — CT602]
79. Why SSB modulation scheme is preferred over DSB, DSB-SC modulation schemes? The total content of an AM signal is 1000W. Determine the power being transmitted if carrier frequency and at each sideband when modulation percentage is 100%. [2078 Chaitra — EX656]
80. Derive the expression of double-tone AM, and draw its line spectrums. Consider a sinusoidal signal  $m(t) = A \cos \omega_m t$  applied to a delta modulator with step size  $\Delta$ . Show that the slope overload distortion will occur if  $> \frac{\Delta}{\omega_m T_s}$ . [2082 Kartik — EX656]
81. Derive the expression of a double-tone AM and draw its line spectrums. [2078 Chaitra — EX656]
82. Derive the time domain and frequency domain expressions of DSB-FC wave of a double-tone (single message-two carriers) AM, and draw their line spectrums. [2082 Chaitra — EX656]
83. Derive the expression for double-tone AM. How DSB is different from SSB signal? [2079 Chaitra — EX656]
84. Derive the expression of double-tone AM, and define BW and modulation indices. For the given modulated signal draw line spectrums:  $s(t) = 50 \cos(2\pi 10^6 t + 60^\circ) - 15 \cos(2\pi 10^6 t - 60^\circ) + 20 \sin(4\pi 10^4 t - 190^\circ) + \sin(2\pi 10^4 t + 190^\circ)$ . [2080 Chaitra — EX656]
85. Write short note on: Double-tone AM. [2078 Bhadra, 2082 Baishakh — CT602]
86. Describe Hilbert Transformation and its properties. Compute the energy and power of unit step signal. [2078 Chaitra — EX656]

87. Hilbert transform define with mathematical expression in frequency domain. [2080 Chaitra — EX652]
88. Write short note on: Hilbert transform and its properties. [2081 Ashwin, 2082 Kartik, 2082 Chaitra — EX656]

### **DSB-FC: Generation (Square Law) and Detection**

89. Explain the generation of DSB FC signal using square law modulator with appropriate diagram and spectrum of transmitted signal. [2080 Chaitra — EX652]

### **DSB-SC: Generation (Linear/Balanced Modulator), Synchronous Detection**

90. What is conventional amplitude modulation? Explain the ring modulator for the generation of double side band amplitude modulation with neat and clear diagram. [2079 Ashwin — EX652]
91. Compare and contrast DSB-SC and SSB. Explain the synchronous demodulation for DSB-SC signal along with its requirements and limitations. [2080 Chaitra — EX652]
92. Explain the effect of phase and frequency error in local oscillator in demodulating DSB and SSB using PLL. [2080 Chaitra — EX652]

### **Overview of SSB, VSB, and ISB**

93. Calculate the power saving for double side band AM signal and single side band AM signal for percentage modulation of 60%. [2079 Ashwin — EX652]
94. Calculate the percentage of power saved in: (i) DSB-SC (ii) SSB-SC as compared to standard AM if the modulation depth is 60%. Describe QAM modulation and prepare constellation diagram of 16-QAM. [2080 Ashwin — EX656]
95. Derive the equation for Single Side-Band modulated signal in terms of Hilbert Transformation of modulating signal  $m(t)$ . Briefly discuss any one method of generating SSB signal. [2079 Chaitra — EX652]
96. Derive the expression for figure of merit for the SSB receiver. Define an efficient constellation diagram of a 32-QAM. [2081 Chaitra — EX656]

### **Phase Locked Loop (PLL), AM Demodulation using PLL**

97. What is PLL? Explain the generation of Narrow Band FM using Armstrong method. Why amplitude limiter is used in limiter discriminator method? [2079 Ashwin — EX652]
98. What is superheterodyne AM receiver? Explain the coherent detection and phase error effect of DSB AM. [2079 Ashwin — EX652]

### **Super-heterodyne AM Receiver**

99. (See superheterodyne AM receiver entry above under PLL, which jointly covers receiver operation.) [2079 Ashwin — EX652]

### 3 Angle Modulation

#### Basic Definitions, Time Domain Expressions for FM and PM

100. The equation of an angle modulated signal is  $v(t) = 12 \sin(10^6 t + 5 \sin 10^4 t)$ . Determine: (a) Carrier frequency (b) Modulating frequency (c) Modulation index (d) Power dissipated in  $100 \Omega$  resistors. [2080 Chaitra — EX652]

#### Time Domain Expression for Single Tone Modulated FM Signals, Spectral Representation

101. A modulated signal is given by  $s(t) = 40 \cos(100.1 \times 10^6 \times 2\pi t + 10 \sin(15 \times 10^3 \times 2\pi t))$ . Determine (i) practical bandwidth (ii) frequency sensitivity (iii) frequency deviation (iv) modulation index. [2079 Ashwin — EX652]
102. A message signal  $m(t) = 10 \cos(2\pi \times 4000t)$  V is used to frequency modulate the carrier signal  $c(t) = 80 \cos(2\pi \times 10^7 t)$  V. Assuming the frequency sensitivity of the frequency modulator to be 400 Hz/Volt, calculate: (a) Peak frequency deviation (b) Modulation index (c) Bandwidth of modulated signal for over 98% of FM power (d) Total modulated signal power dissipated in unit impedance. [2079 Chaitra — EX652]
103. Write short note on: Properties of Bessel function. [2079 Ashwin — EX652]

#### Bandwidth of FM, Carson's Rule, Narrow and Wideband FM

104. Differentiate NBFM and WBFM. Explain the functioning of stereo encoder and decoder in commercial FM with clear diagram. [2079 Ashwin — EX652]

#### Generation of FM: Direct and Indirect

105. Explain the Armstrong's method for Frequency Modulation. [2080 Chaitra — EX656]

#### Demodulation of FM: Non-synchronous and Synchronous

106. (Non-synchronous/synchronous FM demodulation is addressed jointly with PLL/Armstrong questions above under Section 2 and 3.)

#### Stereo FM, Pre-emphasis and De-emphasis Network

107. What is the relation between psdf and Autocorrelation function? Explain the Stereo FM encoder and decoder with spectral diagram. [2078 Chaitra — EX656]
108. With functional block diagrams and spectral details explain the operation of stereo encoder and decoder. [2079 Chaitra — EX652]
109. Explain Pre-emphasis and De-emphasis. [2081 Ashwin — EX656]
110. Why do we need pre-emphasis and de-emphasis circuits in FM? Explain. [2080 Chaitra — EX656]
111. Write short notes on: Pre-emphasis and de-emphasis; Stereo encoder; Filter and oscillator requirement in FDM. [2080 Chaitra — EX652, 2081 Chaitra — EX656]

#### Super-heterodyne Radio Receiver for FM

112. Explain the operation of FM superheterodyne radio receiver. [2080 Chaitra — EX652]

## 4 Pulse Modulation

### Sampling Theorem, Ideal/Practical Sampling, Aliasing, Aperture Effect, Reconstruction

113. Explain the aperture effect during flat-topped sampling. Illustrate the DPCM scheme that overcomes the disadvantages of PCM. A delta modulator system is designed to operate at 5 times the Nyquist rate for a signal having a bandwidth equal to 3kHz bandwidth. Calculate the maximum amplitude of a 2kHz sinusoidal for which the delta modulator does not have slope overload. The given step size is 250 mV. [2078 Chaitra — EX656]
114. State the practical considerations that need to be considered during sampling. Explain natural sampling with appropriate derivation. [2081 Bhadra — EX702]
115. State and prove Nyquist sampling theorem. [2080 Bhadra, 2080 Baishakh — EX702]
116. Briefly explain flat top sampling. Explain flat top sampling with appropriate derivation, and compare its advantage over natural sampling. Find the Nyquist rate and the Nyquist interval for the signal  $x(t) = \sin(800\pi t)/\pi t + 10 \cos 2000t$ . [2081 Baishakh, 2080 Baishakh, 2079 Bhadra — EX702]
117. Explain the merits and demerits of flat-top sampling of a continuous time signal using relevant mathematical expressions. [2081 Baishakh — EX702]

### Fundamentals of PAM, PWM, PPM

118. Explain different types of pulse modulation techniques with necessary diagrams. [2080 Bhadra — CT602]
119. Differentiate between pulse amplitude and pulse position modulation. Explain the generation of pulse width modulation. [2081 Bhadra — EX702]
120. Write short note on: PAM, PWM and PPM. [2079 Bhadra — EX702]

### PCM: Quantization, Quantization Error and Noise

121. Explain the procedure of converting an analog signal to digital. Briefly explain each step involved. [2078 Kartik — CT602]
122. Define PCM technique used to convert analog signal into digital data. In a PCM system, a bandlimited signal with maximum frequency 4 kHz is sampled at the rate of Nyquist frequency. It is quantized into 256 levels and encoded. Calculate: (i) Minimum data rate available at the output of the encoder. (ii) Bandwidth of the channel required to transmit the data through a channel of 15 dB SNR. [2080 Baishakh — CT602]
123. A bandlimited signal with maximum frequency 4 kHz is sampled at the rate of Nyquist frequency. It is quantized into 128 levels and encoded. Calculate: Bandwidth of the channel required to transmit the data through a channel with SNR of 0 dB. [2081 Bhadra — CT602]
124. Describe PCM with a block diagram. [2082 Bhadra — CT602]
125. Define Pulse Code Modulation (PCM). A band-limited signal  $m(t)$  with a bandwidth of 3 kHz is sampled at a rate equal to twice the Nyquist rate. The step size used during the quantization process is 0.5% of the peak amplitude  $m_p$ . Assuming binary encoding, determine the minimum bandwidth of the channel to transmit the encoded binary signal. [2081 Baishakh — EX702]

126. Define quantization. Derive the expression for evaluating signal to quantization noise ratio (SQNR) for linear quantization. [2080 Bhadra — EX702]
127. Show that for a voice transmission, the basic data rate using PCM is 64 kbps. [2080 Ashwin — EX656]

### **SQNR in Uniform Quantization, Non-uniform Quantization, Companding**

128. Why is non-uniform quantization required? Show that for a voice transmission, the basic data rate using PCM is 64 kbps. [2080 Ashwin — EX656]
129. Elaborate on the technique of implementing non-uniform quantization using uniform quantization technique. With a suitable diagram, illustrate the working of  $\mu$  law. [2081 Bhadra — EX702]
130. What is companding? A Delta modulator is used to encode speech signal band-limited to 5 KHz with a sampling frequency of 200 KHz. For maximum signal amplitude of  $A_{max} = 1$ , find: (a) Minimum step size to avoid slope overloading (b) Assuming the speech signal to be sinusoidal, find signal to quantization noise ratio (c) Determine the minimum transmission bandwidth. [2079 Bhadra — EX702]

### **DPCM, DM: Encoder, Decoder, Advantages, Disadvantages, Noise in DM**

131. Illustrate the DPCM scheme that overcomes the disadvantages of PCM. A delta modulator system is designed to operate at 5 times the Nyquist rate for a signal having a bandwidth equal to 3kHz bandwidth. Calculate the maximum amplitude of a 2kHz sinusoidal for which the delta modulator does not have slope overload, given step size 250 mV. [2078 Chaitra — EX656]
132. Compare Pulse Code Modulation (PCM), Differential Pulse Code Modulation (DPCM) and Delta Modulation. Find the Signal to Quantization Noise ratio (SQNR) of Pulse Code Modulation (PCM). [2079 Chaitra — EX656]
133. Compare Differential Pulse Code Modulation (DPCM) and compare with Delta Modulation. Explain E1 TDM-PCM Telephone Hierarchy. [2080 Baishakh — EX702]
134. Derive an expression for the SQNR (in dB) of the delta modulation for a single-tone analog signal. [2081 Baishakh — EX702]
135. Explain the Delta and Adaptive Delta Modulations encoders and decoders with their derivations and diagram. List out their merits and demerits. [2080 Chaitra — EX656]
136. Write short note on: Delta modulation. [2081 Ashwin — EX656]
137. Explain the principles of Adaptive Delta Modulation (ADM). Explain T1 digital carrier system with its multiplexing hierarchy. [2079 Bhadra — EX702]

## **5 Multiplexing Techniques**

### **Multiplexing Fundamentals: FDM, TDM, WDM and Applications**

138. Compare all three types of multiplying technique with the application case of each. [2079 Bhadra — CT602]
139. Define multiplexing? Why statistical TDM is preferred than synchronous TDM? [2080 Bhadra — CT602]

140. Explain the major bandwidth utilization techniques used in data communication. [2080 Baishakh — CT602]
141. Write down the applications of multiplexing. [2081 Baishakh — CT602]
142. Compare digital and analog hierarchies with detailed block diagram showing number of channels and data-rate. [2082 Baishakh — ENCT253, 2083 Baishakh — ENCT253, 2082 Bhadra — ENCT253]
143. Explain the FDM telephone hierarchy. Describe the filter and oscillator requirement in FDM. [2079 Ashwin — EX652]
144. Define FDM. Explain FDM hierarchy used in telephony. [2079 Chaitra — EX652]
145. Briefly explain filter and oscillator requirement in Frequency Division Multiplexing (FDM). Compare packet switching and message switching. [2081 Ashwin — EX656]
146. Write short notes on: Filter and oscillator requirement in FDM. [2080 Chaitra — EX652]
147. Compare TDM and FDM. Compare E1 and T1 hierarchies for voice application. [2079 Chaitra — EX656]
148. Write short note on: FDM vs TDM. [2081 Chaitra, 2082 Chaitra — EX656]
149. Write short note on: Multiplexing vs multiple access. [2081 Ashwin — EX656]

### **T1 and E1 TDM PCM Telephony Hierarchy**

150. Explain T1 and E1 hierarchy with its necessary calculation. [2082 Bhadra — CT602]
151. Write short notes on: E1 and T1 Hierarchies. [2080 Bhadra — CT602]
152. Write short note on: T1 Hierarchy. [2082 Baishakh — CT602]
153. Write short note on: E1 Hierarchy. [2081 Bhadra — CT602]
154. Draw T1 and E1 telephone hierarchy (with FDMA/TDMA differentiation). [2078 Chaitra — EX656]
155. Compare E1 and T1 TDM PCM hierarchy. What is the significance of constellation diagram? [2080 Chaitra — EX656]
156. Write short note on: T1 vs E1 hierarchy. [2081 Ashwin — EX656]
157. Write short notes on: T1 and E1 Hierarchy used in telephony. [2081 Chaitra, 2082 Chaitra — EX656]
158. With the help of the frame diagrams, discuss T1 and E1 hierarchy of TDM-PCM telephony. [2080 Bhadra — EX702]
159. Explain T1 digital carrier system with its multiplexing hierarchy. [2079 Bhadra — EX702]

## Multiple Access Fundamentals: FDMA, TDMA, CDMA, SDMA

160. Explain the "near-far problem" in CDMA. How can it be solved? [2078 Kartik — CT602]
161. Explain the mechanism of frequency Hopping spread spectrum (FHSS). Also, compare FDM and FHSS using suitable time-frequency graph. [2078 Kartik — CT602]
162. Explain how FHSS working principle and explain how CDMA works with example. [2078 Bhadra — CT602]
163. Discuss briefly on Spread Spectrum communication with neat block diagram. Compare all three types of multiplexing technique with the application case of each. [2079 Bhadra — CT602]
164. Determine the CDMA encoded transmitting message for four users using Walsh Hadamard codes, where user A, B, C and D are transmitting 1, 0, silent, 1 respectively. [2079 Bhadra — CT602]
165. Write down the applications of multiplexing. Describe FHSS and DSSS with necessary diagrams. [2081 Baishakh — CT602]
166. Write short note on: Shannon's channel capacity theorem; CDMA. [2081 Baishakh — CT602]
167. What are Walsh codes? Explain with examples. How is it implemented in CDMA systems? [2082 Bhadra — CT602]
168. Demonstrate the operation of CDMA for 8 users among whom four users are sending bit 1, two users are sending bit 0 and remaining two are silent. [2082 Bhadra — ENCT253]
169. Demonstrate the operation of CDMA for 8 users among whom three users are sending bit 1, three users are sending bit 0 and remaining two are silent. [2083 Baishakh — ENCT253]
170. Write short note on: Evolution of mobile communication technologies from 1G to 5G. Write short note on: FHSS and DSSS. [2082 Bhadra, 2083 Baishakh — ENCT253]
171. Differentiate between FDMA and TDMA. [2078 Chaitra — EX656]
172. Construct a (7,4) binary CRC using a generator polynomial  $g(x) = x^3 + x^2 + 1$  with data Vector (1011). Demonstrate how CRC-4 detects two burst errors. [2080 Ashwin — EX656]

## 6 Baseband Digital Data Transmission

### Information Theory, Entropy, Symbol Rates and Data Rates

173. A discrete source is emitting one of the 5 possible symbol per microsec. The probabilities are  $1/2$ ,  $1/4$ ,  $1/8$ ,  $1/16$  and  $1/16$ . Find (i) symbol rate (ii) source entropy, and (iii) information rate. [2082 Bhadra — CT602]
174. What is baud rate? If a communication channel has a bandwidth of 3 kHz and a SNR of 31 dB, what would be the channel capacity using Shannon channel capacity theorem? [2082 Bhadra — CT602]
175. What is source coding different from channel coding? What is the aim of source coding? [2078 Kartik — CT602, 2079 Chaitra — EX656]

176. Differentiate message and information. Relate the message, the entropy and the information. [2080 Bhadra — EX702]
177. Mention how can we minimize error in duo-binary coding. Define information and entropy. Calculate the upper limit of the channel capacity as the bandwidth of the channel (B) tends to infinity. [2081 Baishakh — EX702]

### Shannon-Hartley Channel Capacity Theorem

178. How Nyquist theorem applied for a noiseless channel? Calculate number of discrete signal content in the channel if a channel has a spectrum of 3 to 4 MHz with signal to noise ratio of 24 dB. [2078 Kartik — CT602]
179. State and explain theorem for noisy and noiseless channel. Briefly discuss about the measures that are used to characterize the performance of channel. [2079 Bhadra — CT602]
180. Compare coaxial cable with optical fibre. Evaluate the maximum capacity of a channel through which a signal with power of 10 dBW is transmitted (assume noise power spectral density 5 mW/Hz). [2080 Baishakh — CT602]
181. What are the different propagation methods of radio waves? Explain. What are the implications of the Shannon channel capacity theorem? Derive the expression for maximum channel capacity when the bandwidth of the channel is infinite. [2082 Baishakh — CT602]
182. What is baud rate? If a communication channel has a bandwidth of 3 kHz and a SNR of 31 dB, what would be the channel capacity using Shannon channel capacity theorem? (repeated calculation) [2082 Bhadra — CT602]
183. Write short note on: Shannon-Hartley channel capacity theorem. [2083 Baishakh, 2081 Baishakh — CT602]
184. State and explain Nyquist Channel Capacity Theorem. [2080 Bhadra — EX702]
185. Explain Shannon Hartley channel capacity theorem and its implication and theoretical limits. Show that channel capacity is  $1.44S/N_0$ , when channel bandwidth tends to infinity. [2080 Baishakh — EX702]
186. With a suitable example explain Shannon-Hartley channel capacity theorem. [2080 Chaitra — EX656]
187. Discuss the limitations of Shannon Hartley Channel capacity theorem. [2082 Kartik — EX656]

### Compression Techniques: Shannon-Fano, Huffman Codes

188. Write down the Huffman Algorithm clearly and find an efficient code word and efficiency that can be assign to the symbols using Huffman Algorithm for "Kun Mandir Ma Janchhau Yattri". [2078 Kartik — CT602, 2079 Chaitra — EX656]
189. Encode "Jasta lai tastai dhido lai nistai" using weighted Huffman encoder. Also demonstrate how it is decoded. [2078 Bhadra — CT602]
190. Apply the Huffman coding procedure to determine Entropy, Average Code length, code efficiency and Redundancy for the following message ensemble  $[x] = [x_1, x_2, x_3, x_4, x_5, x_6]$  and  $[p] = [0.30, 0.25, 0.15, 0.12, 0.08, 0.10]$ , Take  $M = 2$ . [2079 Bhadra — CT602]

191. Encode "Machho Machho Bhyaguto" using Huffman Encoding and calculate its transmission efficiency. What would be the code sequence if the recipient has wrongly received the message like "Machho Machho Machho"? [2082 Baishakh — CT602]
192. Encode "SaReGaMaPaDhaNiSa" using Huffman Encoding and calculate its transmission efficiency. What would be the code sequence if the recipient has wrongly received the message like "SaReSaRePaDhaNiSa"? [2081 Bhadra — CT602]
193. Generate the Huffman code of word "Information" and decode to message it properly. [2082 Bhadra — CT602]
194. Encode 'Gen-Z Generation' using Huffman coding technique, and calculate the transmission efficiency. [2083 Baishakh — CT602]
195. Encode "I'm 100% OK." using weighted Huffman Encoder, and find the transmission efficiency. [2082 Bhadra — ENCT253]
196. Encode " $E = \int_{-\infty}^{+\infty} |x(t)|^2 dt$ " using weighted Huffman encoder and calculate entropy. [2082 Chaitra — EX656]
197. Encode "Engineer Ko Betha Arulai Ke Thaha" using Huffman Encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Engineer Ko Betha Ke Ke". [2081 Chaitra — EX656]
198. Encode "Ghumne Mech Mathi Andho Manche" using Huffman coding to determine efficiency redundancy and compare it with entropy. [2081 Baishakh — CT602]
199. Encode "Kun Mandir Ma Janchhau Yatri" using Huffman codes and finds its efficiency. [2079 Chaitra — EX656]
200. A DMS has five symbols with probability of  $p(x_1) = 0.4$ ,  $p(x_2) = 0.19$ ,  $p(x_3) = 0.16$ ,  $p(x_4) = 0.15$  and  $p(x_5) = 0.1$ . Construct the Huffman code for X and calculate its efficiency and compare with fixed Length coding. [2081 Baishakh — EX702]
201. A discrete memoryless source has an alphabet of seven symbols whose probabilities of occurrence are:  $S_0$  to  $S_6$  with probabilities 0.25, 0.25, 0.125, 0.125, 0.125, 0.0625, 0.0625. Determine the Huffman code for above source. [2079 Bhadra — EX702]
202. A DMS has six symbols with probability of  $p(x_1) = 0.3$ ,  $p(x_2) = 0.2$ ,  $p(x_3) = 0.25$ ,  $p(x_4) = 0.12$ ,  $p(x_5) = 0.08$ ,  $p(x_6) = 0.05$ . Construct the Shannon-Fano code for X and calculate its efficiency and code Redundancy. [2080 Baishakh — EX702]
203. Encode "Ma Mare Pani Mero Desh Bachi Rahos" using Shannon-Fano codes and calculate its efficiency. [2080 Ashwin — EX656]
204. Encode "Mississippi is missing" using weighted Huffman code and calculate transmission efficiency. [2080 Chaitra — EX656]
205. Encode 'Phoola Ko Thunga Bahera Gayo Ganga Ko Panima' using Shannon-Fano technique and calculate the transmission efficiency. [2081 Ashwin — EX656]
206. Encode "Rain Rain Go To Spain" using Shannon-Fano encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Rain Rain Go Go Spain". [2082 Kartik — EX656]
207. Encode "Solve  $\{(a+b)(a-b)\}^2$ ." using Shannon-Fano Encoder, and find the transmission efficiency. [2083 Baishakh — ENCT253]

### **Line Coding Schemes: Unipolar, Polar, Bipolar**

208. Define line coding. Explain polar RZ and bipolar AMI line coding scheme with example and compare them. [2078 Bhadra — CT602]
209. What are the basic factors to be considered while line coding? Encode the binary pattern 100000000100001 using B8ZS and HDB3 encoding. [2080 Baishakh — CT602]
210. Explain the importance of Scrambling in the digital data encoding process. Encode the binary pattern 101000011101 using polar RZ, NRZ-I, AMI and Manchester line coding techniques. [2081 Baishakh — CT602]
211. Define unipolar, polar, bi-polar, unipolar RZ and Manchester line codes. [2079 Chaitra — EX652]
212. What is modulation needed in communication system? Encode the bit sequence 01100000001100000110 using B8ZS and HDB3 scheme. [2082 Bhadra — CT602]

### **RZ, NRZ, AMI, Manchester, Differential Manchester, B8ZS, HDB3**

213. Encode 111000000000000011 using B8ZS and HDB3 encoders. [2078 Bhadra — CT602]
214. Encode the bit stream 101110001010 using NRZ-L, NRZ-I, Manchester, Bipolar AMI. [2079 Bhadra — CT602]
215. Encode the binary pattern 100000000101 using B8ZS and HDB3 encoding. [2080 Baishakh — CT602]
216. Encode the given bit sequence 101000000000101 using any suitable two of polar and two of bipolar encoders. [2081 Bhadra — CT602]
217. Encode the given bit sequence 101000001 using Unipolar RZ, Polar RZ, AMI and HDB3 encoders. [2082 Baishakh — CT602]
218. Encode 100000000101 using Polar RZ and B8ZS coding schemes. [2083 Baishakh — CT602]
219. Represent 100111010 using following encoders: (a) Polar RZ (b) Bipolar NRZ (c) AMI (d) Manchester. [2078 Chaitra — EX656]
220. Given the binary sequence 1101010111 represent in Unipolar RZ, Bipolar NRZ, Polar NRZ and Manchester encoders. Explain communication impairments with examples. [2079 Chaitra — EX656]
221. Given the binary sequence 1001101101, represent in unipolar NRZ, polar RZ, AMI, and Manchester codes. [2081 Bhadra — EX702]
222. Represent the given binary sequence 110011000001 in unipolar RZ, Manchester and HDB3 encoders. [2080 Chaitra — EX656]
223. Represent 100111010 using Unipolar RZ, Bipolar NRZ, Manchester and AMI encoding technique. [2080 Ashwin — EX656]
224. Encode the given bit sequence 101000001 using Unipolar RZ, AMI and HDB3 encoders. [2081 Ashwin — EX656]
225. Consider that the bit sequence 100000000001101 is to be transmitted. Draw the resulting waveform if the sequence is transmitted using Polar RZ, Differential Manchester, HDB3 and B8ZS, and compare their bandwidths. [2081 Chaitra — EX656]

226. Draw the timing diagram of Unipolar RZ, Polar NRZ, AMI and B8ZS for a binary sequence of 101100001000000000. [2082 Kartik — EX656]
227. By assuming suitable bit streams, demonstrate how HDB3 and B8ZS line encoders work. [2082 Chaitra — EX656]
228. Given the binary sequence 1101010111 represent in polar NRZ, polar RZ, Manchester and AMI codes. [2081 Baishakh — EX702]
229. Draw the waveform for the bit sequence 1000000000101 using each one suitable unipolar, polar and bipolar encoders. [2082 Bhadra, 2083 Baishakh — ENCT253]
230. Represent the given binary sequence 1001101101 in unipolar NRZ, polar RZ, AMI, and Manchester codes. [2081 Bhadra — EX702]

### ISI, Nyquist Criteria, Pulse Shaping for Zero ISI

231. Define  $BW$ ,  $SNR$ ,  $SQNR$ ,  $BER$ ,  $ISI$  parameters. [2081 Ashwin — EX656]
232. Define Inter Symbol Interference. Elaborate Nyquist pulse shaping criterion for mitigation of ISI. [2081 Bhadra — EX702]
233. Define Inter Symbol Interference (ISI). State Nyquist conditions for zero ISI. What are the physical constraints in implementing Nyquist pulse shaping criteria? [2080 Bhadra — EX702]
234. Illustrate the methods to reduce Intersymbol Interference. Explain modified Duobinary coding technique with precoder, illustrating with input sequence 01001101. [2079 Bhadra — EX702]

## 7 Digital Modulation Techniques

### Binary Digital Modulation (ASK, FSK, PSK)

235. Differentiate ASK from PSK. Prepare a constellation diagram of 16-QAM. [2083 Baishakh — CT602]
236. Write short note on: ASK, PSK and FSK. [2079 Ashwin — EX652]
237. Describe ASK, PSK and FSK with signal diagrams. [2083 Baishakh — ENCT253]
238. Explain the block diagram of generation and detection of BPSK. [2082 Bhadra — CT602]
239. Explain BPSK modulation technique with its relevant diagram and signal space diagram. [2078 Chaitra — EX656]
240. Briefly explain generation and detection of Phase Shift Keying (PSK) with necessary illustration. Discuss Gaussian Minimum Shift Keying (GMSK) with its advantage. [2081 Baishakh — EX702]
241. Explain the modulation scheme for MSK with appropriate example. [2080 Chaitra — EX656, 2082 Chaitra — EX656]
242. With the help of neat circuit diagram, explain the operation of the square law detector. [2081 Ashwin — EX656]
243. Explain the block diagram of generation and detection of BPSK. [2082 Bhadra — CT602]

244. Compare A.M and F.M in terms of power efficiency and system complexity. Calculate the error probability of coherent ASK. [2080 Baishakh — EX702]
245. Illustrate MSK signal for a message of 1011010. [2081 Ashwin — EX656]

### **QPSK Generation, Properties, Constellation Diagram, Detection**

246. What are the significances of multilevel modulation? Explain QPSK with its transformation as well as a receiver block diagram. [2078 Chaitra — EX656]
247. Explain the Modulator, Demodulator, and Signal Space Diagram for QPSK modulation with relevant derivations. [2080 Baishakh — EX702]

### **M-ary Modulation Techniques, M-PSK versus M-QAM**

248. Explain briefly about 8-QAM with constellation diagram. [2080 Bhadra — CT602]
249. Explain 16-QAM with constellation diagram. [2081 Baishakh — CT602]
250. Prepare a constellation diagram of 32-QAM. [2081 Ashwin — EX656]
251. Prepare a constellation diagram of 32-QAM (efficient constellation). [2081 Chaitra — EX656]
252. Define an efficient constellation diagram of a 32-QAM. [2082 Kartik — EX656]
253. Write short note on: 32-QAM. [2082 Baishakh — CT602, 2082 Kartik — EX656]
254. Explain QAM modulation and demodulation with its required diagram. [2078 Chaitra — EX656]
255. Explain the working of Quadrature Amplitude Modulation (QAM) with appropriate modulation, demodulation block, and constellation diagram. [2081 Bhadra — EX702]
256. What are the significance of constellation diagram? Compare E1 and T1 TDM PCM hierarchy. [2080 Chaitra — EX656]
257. Write short note on: M-ary baseband communication system. [2079 Bhadra — EX702]

## **8 Error Detection and Correction Coding**

### **Hamming Weight, Hamming Distance, Code Vectors, Constraint Length, Code Rate, Syndromes**

258. Define Hamming distance and Minimum hamming distance with an example. What are the error detection and error correction capabilities of block codes? Explain. [2080 Bhadra — CT602]
259. What is the significance of ( $d_{min}$ ) minimum hamming distance? [2081 Bhadra — EX702]
260. Write short note on: Error detection and correction capability of a codeword. [2081 Bhadra — EX702]

## Error Detection Codes: Checksum, CRC

261. Explain the operation of CRC-4 with example of error detection. [2078 Kartik — CT602]
262. Demonstrate how checksum is used to detect errors while sending a data word of 12, 15, 10, 5, 2. [2078 Bhadra — CT602]
263. With suitable example, demonstrate how checksum detects an error in data communication. [2083 Baishakh — CT602]
264. Explain CRC-4 generator and decoder with example for both no-error and with-error cases. [2081 Baishakh — CT602]
265. Construct a (7,4) binary CRC using a generator polynomial  $g(x) = x^3 + x^2 + 1$  with data Vector (1011). Demonstrate how CRC-4 detects two burst errors. [2080 Ashwin — EX656]
266. With suitable examples, introduce how CRC-3 and LBC work to correct a single bit error. [2082 Kartik — EX656]
267. Write short note on: CRC-4. [2082 Bhadra — ENCT253]
268. Write short note on: CRC-3. [2082 Baishakh — CT602]
269. Write short note on: Checksum. [2083 Baishakh — ENCT253]

## Error Correction Codes: Linear Block Codes, Hamming Codes

270. What are linear block codes? Design a code word of a C(8,4) block code with any suitable generation matrix. [2078 Bhadra — CT602]
271. For a (6,3) code, the generator matrix is

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Find (i) All corresponding code vectors (ii) Minimum Hamming distance (iii) Verify that this code is a single error correcting code (iv) Parity check matrix (v) Determine transmitted codeword if received code is 100011. [2079 Bhadra — CT602]

272. Obtain all code words for a (6,3) block code from the systematic generator matrix given below:

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

[2080 Bhadra — CT602]

273. A generation matrix of C(6,3) linear block code is given as

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

(a) Find code word for message 110. (b) Decode the received sequence 110110. [2082 Baishakh — CT602]

274. A (6,3) error control code has the following parity check matrix

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Determine the generator matrix G (b) Generate a codeword for message 110 (c) Check whether the received code word 110111 as has error or not. [2081 Chaitra — EX656]
275. Determine and correct the error bit in the even parity received Hamming codeword of 1110011. [2082 Bhadra — ENCT253]
276. Using any example of a 3x3 generation matrix, demonstrate how LBC corrects a single bit error. [2083 Baishakh — ENCT253]
277. With suitable assumption for a code word of C(7,4), demonstrate how Hamming Code detects an error. [2081 Ashwin — EX656]
278. Generate a hamming code for 10101100 using even parities, and demonstrate how it corrects a single bit error. [2082 Chaitra — EX656]
279. Prove that (4,3) even parity code is a linear block code and (4,3) odd parity code is not a linear block code. [2081 Baishakh — EX702]
280. Write short note on: Hamming codes. [2078 Bhadra — CT602, 2083 Baishakh — CT602]
281. Write short note on: Linear Block Coding. [2083 Baishakh — CT602]
282. Check whether the received code word 110111 as has error or not. Encode "Rain Rain Go To Spain" using Shannon-Fano encoder and calculate the transmission efficiency. Find the codeword if the message is received as "Rain Rain Go Go Spain". [2082 Kartik — EX656]

### Cyclic Codes (Generator Polynomial, Parity-Check Polynomial)

283. Under what conditions does a linear code become a cyclic code? Explain with the help of an example. [2078 Kartik — CT602]
284. Determine systematic and non-systematic code vector for a (7,4) cyclic hamming code for message vector {1011} with generator matrix  $g(x) = 1 + x + x^3$ . [2078 Chaitra — EX656]
285. What is binary Cyclic code? Construct a (7,4) binary Cyclic Code using a generator polynomial  $g(x) = x^3 + x^2 + 1$  with data Vector (1011). [2080 Baishakh — EX702]
286. Determine the systematic and non-systematic code vector for a (7,4) cyclic code for message vector  $(m_0, m_1, m_2, m_3) = (1101)$  for a given generator polynomial  $g(x) = 1 + x + x^2$ . [2081 Bhadra — EX702]
287. Write short note on: A (7,4) polynomial cyclic Hamming code given by  $g(x) = 1 + x + x^3$ . [2082 Kartik — EX656]

## 9 Noise in Communication Systems

### Definition, White Noise, AWGN Channel, PSDF, Autocorrelation of White Noise

288. List out any three properties of autocorrelation (AC) function. Mention the autocorrelation function of white noise. [2078 Chaitra — EX656]

289. Define the power spectral density function and express its relation with the autocorrelation function. Check whether  $y(t) = \log\{x(t)\}$  is a linear or non-linear system. [2082 Kartik — EX656]
290. State Parseval's Theorem. Explain the relation between power spectral density function and autocorrelation function with the example of white noise. [2080 Chaitra — EX652]
291. Write short note on: Band limited white noise vs thermal noise. [2081 Ashwin — EX656]
292. Write short note on: PSD of white noise. [2082 Kartik, 2082 Chaitra — EX656]
293. Write short note on: Autocorrelation function and its properties. [2079 Chaitra — EX652]

### **Ideal Low-pass and RC Filtering of White Noise, Noise Equivalent Bandwidth**

294. Define Noise Equivalent bandwidth. Find noise equivalent bandwidth of the first order RC low pass filter. [2080 Baishakh — EX702]
295. Explain Noise Equivalent Bandwidth represents of a filter. Derive the impulse response of the optimum detector in the presence of additive white noise. [2081 Bhadra — EX702]

### **Optimum Detection, Matched Filter, Realization (Time Correlators)**

296. What do you mean by optimum detector? Show that the impulse response of the matched filter is reverse delayed version of the input signal. [2079 Chaitra — EX656]
297. Define matched filter. Find the impulse response of optimum detector in the presence of additive white noise. [2081 Ashwin — EX656]
298. Derive the expression for the impulse response of a matched filter for an AWGN channel. [2080 Bhadra, 2082 Kartik, 2082 Chaitra — EX656]
299. Write short note on: Impulse response of a matched filter for an AWGN channel. [2082 Kartik — EX656]
300. What is Ergodic Stochastic Process? Derive the expression of impulse response of the matched filter. [2081 Baishakh — EX702]
301. Show that the output signal of LTI (linear time invariant) system is also a wide sense stationary process if the input to it is also a WSSP. [2081 Baishakh — EX702]
302. Define optimum detector and find the impulse response of optimum detector in the presence of additive white noise. Briefly explain Hilbert Transform. [2080 Chaitra — EX656]

### **Overview of Error Probability Function in Digital Communication (ASK, FSK, PSK)**

303. Evaluate the error probability in a binary communication system with appropriate expression. [2081 Bhadra — EX702]
304. Derive the expression for evaluating error probability in Binary baseband system. [2080 Bhadra — EX702]
305. Evaluate the noise performance of the coherent detection of DSB-FC using necessary derivations. Use the result to evaluate the noise performance of the signal-tone DSB-FC modulation with 100% modulation. [2081 Baishakh — EX702]

306. Derive the bit error probability for the coherent binary FSK system. [2081 Baishakh — EX702]
307. Derive the expression for evaluating error probability in binary communication system. What is threshold effect in FM? What are the techniques for the mitigation of threshold effect in FM? [2079 Bhadra — EX702]
308. Write short note on: Threshold effect in WBFM. [2080 Bhadra — EX702]

## A Out-of-Syllabus Questions

309. Write down the two basic approaches commonly used to packet switching. [2079 Bhadra — CT602]
310. Differentiate between circuit switching and Packet switching used in computer networks with an example. [2080 Baishakh — CT602]
311. Write short notes on: Packet switching versus message switching. [2078 Bhadra — CT602]
312. Write short notes on: X.25 protocol. [2078 Bhadra — CT602]
313. Write short notes on: Datagram packet switching vs Virtual circuit packet switching. [2080 Bhadra — CT602]
314. Write short notes on: Modes of operation of optical fiber. [2080 Bhadra — CT602]
315. a) Explain the different modes of data transmission in optical fiber. List out and mention the protocols of different categories of twisted pair cables. [2081 Bhadra — CT602]
316. Write short note on: Difference between circuit switching and packet switching. [2081 Baishakh — CT602]
317. Explain ATM with its frame format. [2082 Bhadra — CT602]
318. Compare optical fiber with UTP. [2083 Baishakh — CT602]
319. Write short notes on: Packet Switching. [2083 Baishakh — CT602]
320. Compare coaxial cable, twisted pair, and optical fiber based on bandwidth, cost, and use case. [2082 Bhadra, 2083 Baishakh — ENCT253]
321. Write short note on: LAN vs WAN. [2082 Bhadra — ENCT253]
322. Write short note on: LAN vs MAN. [2083 Baishakh — ENCT253]
323. Write short note on: Packet switching vs circuit switching. [2083 Baishakh — ENCT253]
324. Write short note on: Evolution of mobile communication technologies from 1G to 5G. [2082 Bhadra, 2083 Baishakh — ENCT253]
325. Explain the concept of convolutional code with the help of a state-transition diagram. [2078 Kartik — CT602]
326. Write short notes on solutions: A (7,4) polynomial cyclic Hamming code given by  $g(x) = 1 + x + x^3$  (convolutional-adjacent context); Single bit error correction using Viterbi decoder. [2082 Kartik — EX656]
327. With suitable example demonstrate how convolutional code corrects errors. [2081 Baishakh — CT602]
328. Design a convolutional encoder having code-rate of  $1/2$ . Also, draw the code-tree and trellis diagram for the same assuming any three-bit input. [2079 Chaitra — EX656]
329. State/tree and trellis diagrams ( $1/2$  code rate and 4 states) using a suitable example. [2080 Bhadra — CT602]
330. With a suitable example of input bit stream show State Diagram, Code Tree and Trellis Diagram. Demonstrate how a convolutional code corrects a single error. [2081 Ashwin — EX656]

331. Why convolution coder is better than block coder? Determine systematic and non-systematic code vector for a (7,4) cyclic hamming code for message vector {1011} with generator matrix  $g(x) = 1 + x + x^3$ . [2078 Chaitra — EX656]
332. Prepare Code Tree, State Space and Trellis diagrams of 1/2 rate convolutional channel coding for a message bit of 1101; and demonstrate how Viterbi Decoding corrects a single bit error. [2081 Bhadra — CT602]
333. Differentiate error-detection and error-correction. Design a convolutional encoder having code-rate of 1/2. Also, draw the code-tree and trellis diagram for the same assuming any three-bit input. [2079 Chaitra — EX656]
334. Determine the encoded sequence for the following input message  $(m_0, m_1, m_2, m_3, m_4) = (10011)$  using convolutional encoder having shift registers with three flip flops. Draw state and trellis diagrams. [2080 Chaitra — EX656]
335. Write short note on: Trellis diagram of a message 1010 using a  $\frac{1}{2}$  rate 3-state memory block convolutional encoder. [2081 Chaitra — EX656]
336. Write short note on: Trellis diagram with example. [2082 Chaitra — EX656]
337. Write short note on: Convolutional coder. [2079 Bhadra — EX702]
338. Write short note on: Convolutional Coding. [2080 Bhadra — EX702]
339. Write short note on: Convolutional Encoding. [2082 Baishakh — CT602]
340. Differentiate convolutional code and block code. [2080 Baishakh — CT602]
341. Determine the systematic and non-systematic code vector for a (7,4) cyclic code for message vector  $(m_0, m_1, m_2, m_3) = (1101)$  for a given generator polynomial  $g(x) = 1 + x + x^2$  (Communication System II, convolutional-adjacent course context). [2081 Bhadra — EX702]
342. Write short note on: Linear prediction theory. [2081 Bhadra — EX702]
343. Write short note on: The eye diagram. [2081 Bhadra — EX702]